Data Storage Layer

GROUP 3

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Data Storage Layer

Stores data for analysis

01

Ensures data is organized

Data from different sources is stored in a structured and consistent way, making it easier to manage, access, and understand.

02

Supports fast queries & analysis

Proper storage and indexing allow systems to retrieve data quickly, enabling efficient analysis and faster insights.

03

Helps decision-making

Well-organized and reliable data provides accurate insights, helping businesses make informed and better decisions

Data Warehouse

Definition

Stores structured & processed data in an organized way for analysis and reporting.

Key Characteristics

- Organized in tables
- Stores historical data
- High data quality (cleaned data)

Purpose

- Used for reporting & analysis
- Supports decision-making

Data Lake

Definition

Stores raw data (all formats)

Key Characteristics

- Structured + unstructured data
- Highly scalable
- Flexible storage

Purpose

- Used for big data & advanced analytics
- Suitable for data scientists

Data Marts

Definition

A subset of Data Warehouse

Key Characteristics

- Department-Specific
- Structured
- Faster Performance

Purpose

- Provides quick access
- Empowers departments to manage their own reporting
- Reduces the search complexity

Data Lakehouse

Definition

A modern architecture that combines the flexibility of Data Lake and structure of a Data Warehouse.

Key Characteristics

- Unified Platform
- ACID Transactions
- Low-Cost Storage

Purpose

- Eliminates Data Silos
- Simplifies the data pipeline

Comparison Between 4 Types of Data Storage

	Data Warehouse	Data Lake	Data Mart	Data Lakehouse
Structure	Structured (tables, schema)	Raw (structured + unstructured)	Structured (subset of warehouse)	Hybrid (structured + raw)
Scalability	Medium (costly to scale)	Very high (big data support)	Low to medium	High (scalable like lake)
Use Case	Reporting, charts	Big data, AI, ML	Department analysis (HR, Sales)	Unified analytics & BI

How Data is Organized ?

Data organized through processes like:

01 **ETL Process**

Extract, Transform, Load

02 **Schema Design**

Tables, rows, columns, organized structure

03 **Metadata**

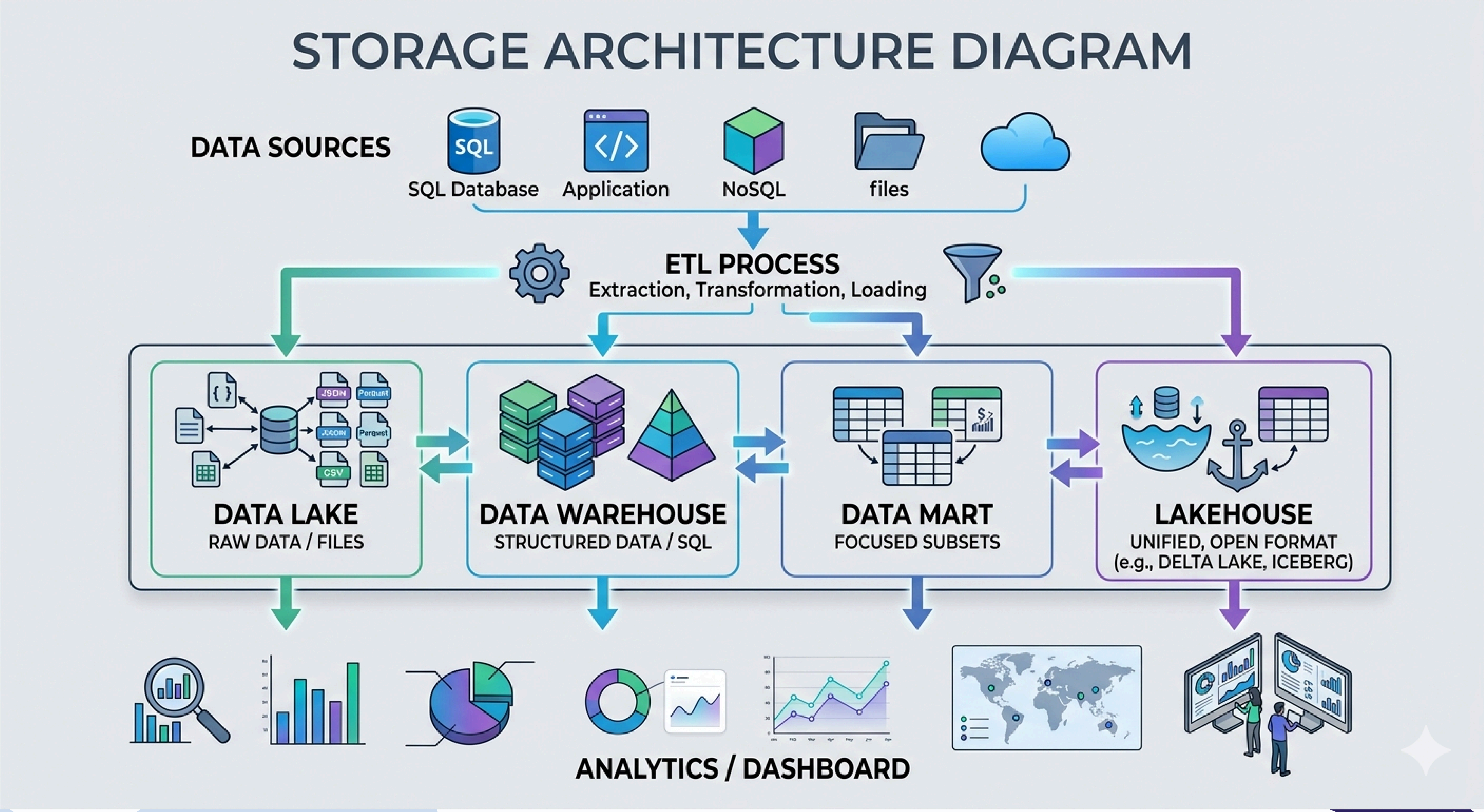
“Data about data”

REAL-WORLD EXAMPLE

Data Lake → stores raw user activity (watch history, clicks)

Data Warehouse → stores processed data (user preferences)

Data Mart → marketing team uses for recommendations


STORAGE ARCHITECTURE DIAGRAM
DATA SOURCES
SQL
SQL Database
Application
NoSQL
files
ETL PROCESS
Extraction, Transformation, Loading
JSON
Parquet
CSV
DATA LAKE
RAW DATA / FILES
DATA WAREHOUSE
STRUCTURED DATA / SQL
DATA MART
FOCUSED SUBSETS
LAKEHOUSE
UNIFIED, OPEN FORMAT
(e.g., DELTA LAKE, ICEBERG)
ANALYTICS / DASHBOARD

THE END

THANK YOU